Lesson A: Watertight Bulkheads

Summary of Lesson

Objectives:

1. To be able to describe the function and operation of bulkheads
2. To investigate the effectiveness of bulkheads
3. To evaluate the effectiveness of bulkheads through experimentation.

This lesson introduces the concepts of bulkheads to students and how they work through practical experimentation. Students will be able to 'select, plan and carry out the most appropriate types of scientific enquiries to test predictions' and finally to 'use appropriate techniques, apparatus, and materials during laboratory work, paying attention to health and safety'. They will also be able to record their results and test come to their own conclusions at the end. This lesson can fit in, if not possible with the coinciding history lesson due to curriculum restraints, with a unit on forces or density (floating and sinking) and can be followed up using "Lesson B: Keeping Afloat".

National Curriculum Links

This lesson focuses on the knowledge and conceptual understanding through:

- practical enquiry into the nature, processes and methods of science through different types of science enquiries that help them to answer scientific questions about the world around them.

By focusing on the sinking of the Titanic this gives students the context of bulkheads in a historical event, but also allows them to explain the sinking of the Titanic by describing:

- associated processes and key characteristics in common language, and [using] technical terminology accurately and precisely.

Equipment

- Lesson PowerPoint
- A large cardboard drinks carton (e.g., 1 litre juice carton)
- Scissors
- Ruler
- Stopwatch
- Waterproof tape
- Two large plastic storage boxes (larger than 10 litre) filled with tap water (to simulate water tanks)
- Resealable waterproof bag
- White clay or play-dough

Lesson Guidance

Step 1: Ask students to write down what they know about the Titanic. If completing the Science lessons alongside the History lessons, please see the resources of these to identify what prior knowledge they should be able to recall. Collate ideas as a class regarding the sinking of the Titanic and question them as to why they think the Titanic sank.

Go through the background detail of the Titanic and explain the concepts of the Watertight Bulkheads and how they function. You may wish to give students a copy of the diagram to annotate with notes.

Step 2: The practical experiment

Ask students to:

1. Follow the instructions on the PowerPoint to build two identical model ships: one without a bulkhead, and one with a bulkhead.
2. Place equal weights (clay in resealable bag) inside both ships to simulate the cargo and passengers. Ensure the weight is evenly distributed.
3. Float the ships in their respective water tanks and observe if they remain stable and at the same water level.
4. Using a pencil, poke a small hole at the same height (about 1.5 cm from the bottom) on both ships to simulate a breach in the hull.
5. Observe and time how long it takes for each ship to begin to sink. Observe and record what happens in each ship. Repeat this process three times.

Step 3: Ask students to feedback their results and findings from the practical regarding the time it took with/without a bulkhead. Students can record these conclusions or discuss as a class. Explain what happened to the bulkheads on Titanic.

Teaching Notes

The ship with a bulkhead will flood more slowly (only the compartment with the hole) and not sink completely. This can be related to the purpose of bulkheads: to separate compartments, maintain buoyancy and control damage. If the observed result is not as expected, check the sealing of the bulkhead, the weight and balance of the clay and the size of the hole.





